

Programming SQL Server 2019

Table constraints

Topics

- Primary key constraint
 - What is it?
 - Applying primary key to a table during table creation
 - At column-level
 - At table-level
 - Dropping primary key
 - Adding primary key to an existing table
- Unique key constraint
 - What is it?
 - Applying Unique constraint

Homework

- A. Create a table 'Equipments' with columns as below

Column Name	Data Type	Allow Nulls	Constraint
id	Char	No	Primary Key, 5-character in length, follows a pattern – 3 uppercase letters followed by two digits ex. TBX98, DMT88
type	Char	No	Value must be one of Lifter, Digger or Mover
price	Money	No	Value always > 0
mindeliverydelay	Int	No	

- B. Add check constraint to mindeliverydelay column based on the rules that mindeliverydelay value must not less than 20
- C. Show the constraints applied to the table

Important points

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Summary

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Primary key constraint

- Primary key constraint makes each row of a table uniquely identifiable
- It enforces entity integrity
- Primary key does not allow duplicate values -two rows in a table cannot have the same primary key
- A primary key can be simple - based on one column
- A primary key can be compound - based on more than one column
- The primary key column has restrictions:
 - A table can have only one primary key
 - All the columns participating in primary key must be NOT NULL, even if you do not specify it

Defining primary key at table design

- Primary can be defined at column level
- Syntax is
`column-name data-type NOT NULL
CONSTRAINT constraint-name PRIMARY KEY`
- CONSTRAINT constraint-name part is optional, you can write
`column-name data-type NOT NULL PRIMARY KEY`
- Primary key can be defined at table level after all column definitions
- Syntax is
`CONSTRAINT constraint-name
PRIMARY KEY (column-name)`

Creating simple primary key

Try this

```
CREATE TABLE trainee
(
    traineeid CHAR(7) PRIMARY KEY,
    traineeName VARCHAR(30) NOT NULL,
    email VARCHAR(50)
)
GO
EXEC sp_helpconstraint 'trainee'
GO
```

- The output is

Object Name
Trainee

constraint_type	constraint_name	..	constraint_keys
PRIMARY KEY (clustered)	PK__trainee__2F0D86AB0F975522	..	traineeid

- traineeid is primary key, and that's why
 - it cannot be null
 - it does not allow duplicates

Try this

- The following will fail, as primary key is not defined
`INSERT INTO trainee (traineeName, email)
VALUES ('Trainee 1', 'some account')
GO`
- Error message would be like below
Cannot insert the value NULL into column 'traineeid', table 'sqlstore.dbo.trainee'; column does not allow nulls. INSERT fails. The statement has been terminated.
- The following is Okay
`INSERT INTO trainee
VALUES ('108967', 'Trainee 1', 'some account')
GO
SELECT * FROM trainee`

GO

- The output is

traineeid	traineeName	email
108967	Trainee 1	some account

- The following will fail, since 108967 traineeid already in in the table in a row
`INSERT INTO trainee
VALUES ('108967', 'Trainee 2', 'trainee@gmail.com')
GO`
- The error message should be like below
Violation of PRIMARY KEY constraint 'PK__trainee__2F0D86AB1367E606'. Cannot insert duplicate key in object 'dbo.trainee'. The duplicate key value is (108967). The statement has been terminated.

Creating compound primary key

- Since a compound primary key is based on more than one column, it must be defined at table level

Try this

```
CREATE TABLE attendance
(
    traineeid CHAR(7) NOT NULL,
    [date] DATE NOT NULL,
    intime TIME NOT NULL,
    outtime TIME,
    CONSTRAINT PK_ATTENDANCE  
PRIMARY KEY (traineeid, [date])
)
GO
```

- traineeid, date pair is primary key – this pair must be unique in the table, that means, there cannot be two attendance record for a trainee on a given date
- The following is okay
`INSERT INTO attendance
VALUES ('108967', '2014-10-21', '08:57', '13:03')
GO
SELECT * FROM attendance
GO`
- The output is

traineeid	date	intime	outtime
108967	2014-10-21	08:57:00.0..	13:03:00.0..

- The following will fail, since a record for 108967 traineeid on 2014-10-21 exists
`INSERT INTO attendance
VALUES('108967', '2014-10-21', '08:43', '13:11')
GO`
- The error message should be like below
Violation of PRIMARY KEY constraint 'PK_ATTENDANCE'. Cannot insert duplicate key in object 'dbo.attendance'. The duplicate key value is (108967 , 2014-10-21). The statement has been terminated.

Dropping primary key

- Syntax:
`ALTER TABLE table-name
DROP CONSTRAINT constraint-name`

Try this

```
ALTER TABLE attendance  
DROP CONSTRAINT PK_ATTENDANCE  
GO
```

Adding primary key by table alteration

- Syntax:
`ALTER TABLE table-name
ADD CONSTRAINT constraint-name`

PRIMARY KEY (column-name)

Try this

```
ALTER TABLE attendance
ADD CONSTRAINT PK_ATTENDANCE
PRIMARY KEY (traineeid, [date])
GO
```

- See the constraints
EXEC sp_helpconstraint 'attendance'
GO
- You should see the primary key constraint in the output

Unique constraint

- Unique constraint like primary key enforces uniqueness of rows
- A table can have only one primary key, but multiple unique constraints
- A Unique key can be created on NULL columns but these columns can have only one NULL value

Defining unique key

- Syntax:
- Unique key be defined at column level
column-name data-type NULL/NOT NULL
CONSTRAINT constraint-name UNIQUE
- Unique key, too, can be defined at table level

Try this

```
CREATE TABLE Users
(
    userid INT NOT NULL IDENTITY PRIMARY KEY,
    username VARCHAR(50) NOT NULL UNIQUE,
    Email VARCHAR(50) UNIQUE
)
GO
```

→userid is primary key
→username unique, two users cannot have the same username
→email allows null and unique

- Let's test it
- The following will work
INSERT INTO Users (username, email)
VALUES ('habib', 'domain_habib@yahoo.com')
GO
- The following fails
INSERT INTO Users (username, email)
VALUES ('habib', 'domain_habib@hotmail.com')
GO
- Error message will be like below
Violation of UNIQUE KEY constraint
'UQ_Users_F3DBC57220C1E124'. Cannot insert
duplicate key in object 'dbo.Users'. The
duplicate key value is (habib).
The statement has been terminated.
- The following also fails
INSERT INTO Users (username, email)
VALUES ('habib1', 'domain_habib@yahoo.com')
GO
- Error message will be like below
Violation of UNIQUE KEY constraint
'UQ_Users_A9D105341DE57479'. Cannot insert
duplicate key in object 'dbo.Users'. The
duplicate key value is
(domain_habib@yahoo.com).
The statement has been terminated.
- You must use different username and email for every user record